

# Is It Really Worth to Peer At IXPs?

## A Comparative Study

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**Abstract**—Internet Exchange Points (IXPs) play a crucial role in the Internet ecosystem. However, existing literature fails in quantitatively assessing the advantage for an Internet Service Provider (ISP) to peer at an IXP. We give a contribution to bridge such a gap by collaborating with three medium-sized ISPs in Italy to compare key performance indicators (round-trip delay, hop count, packet loss, and jitter) as measured from several vantage points in presence and absence of IXP peerings. Our findings are that IXP-based paths exhibit better and more stable performance, whereas avoiding IXPs introduces performance deterioration and higher variability. Moreover, our measurements confirm that IXP-based paths tend to preserve the locality of traffic.

### I. INTRODUCTION

*Internet eXchange Points* (IXPs) are infrastructures used by *Internet Service Providers* (ISPs) to exchange traffic between their *Autonomous Systems* (ASes). An IXP allows ISPs to interconnect their ASes directly, i.e. to establish *peerings* between them, rather than through third-party networks (*upstreams*). IXPs play a crucial role in the development of the Internet, encouraging ISPs to create a dense network of interconnections at low cost. Some of them (e.g., DE-CIX, AMS-IX, and LINX) have a throughput of many Tbit/sec. and are some of the most important building blocks of today's Internet [1].

Despite the importance of IXPs, scientific literature lacks extensive studies that show the impact of peering at IXPs on common network key metrics. In [2], e.g., the authors claim that having peerings at an IXP is efficient in terms of performance, but their assumption is not backed by real data. The main limitation lies in the fact that thorough studies need to directly compare the performance of traffic flowing through an IXP as opposed to that through upstream providers.

In our work we make a first attempt to fill this gap by collaborating with three medium-size ISPs, in order to actively control their BGP announcements and force the traffic to take specific routes for useful comparison. More in detail, we perform experiments in which network paths between two ASes either traverse IXPs or rely only on upstream providers. Such experiments help us determine to which degree IXPs are actually beneficial for involved peers. We perform measurements regarding the following network metrics: *round-trip time* (rtt), *hop count*, *packet-loss*, and *jitter*. Our study is focused on Italy, meaning that both the source ASes and the target ASes of our measurements are located in Italy. We concentrate our attention on the two most famous Italian IXPs: MIX and NaMeX. Our study stems from recent news [3] on major ISPs canceling their peerings at IXPs (*de-peering*). In several occasions they justified such decision in terms of more efficient

handling of IP traffic and improvement of the QoS. As a second topic of interest, we also study whether IXPs are effective in preserving traffic locality, by checking which countries are traversed to reach frequently visited Italian destinations from Italian sources. This is a security issue of concern for many nations. In particular, given the recent espionage cases of network traffic [4], a nation might be interested to see if its own citizens, to reach Internet services that are considered critical, have to cross ISPs of different countries or even of different continents [5].

Our paper is organized as follows. Section II discusses the state of the art in the analysis of IXPs. In Section III we describe the different experiments that we conducted to assess the role of IXPs on the Italian Internet ecosystem. Section IV details the results of our analysis. Finally, Section V presents our conclusions and hints for future work.

### II. RELATED WORK

A survey by Chatzis et al. [2] lists three main reasons for an ISP to establish peerings at an IXP: saving money, improving the QoS, and meeting the requirements of big players (e.g. Google, Netflix) that incentivize other networks to connect at IXPs to obtain peerings with them. However, no experiments have been performed by the authors to support the claim on the QoS. The work also examines the diversity of IXPs around the world, highlighting the non-profit nature and the potential for innovation of European Exchange Points as opposed to standard commercial offers in North America. [6] analyses data on the peerings at Slovak IX, observing that many of them carry only little traffic and therefore their existence would not be economically viable outside of an IXP. Recent research work [1], [7] proves that large IXPs host thousands of peerings between heterogeneous network that carry petabytes of traffic on a daily basis. Further, the analysis of traffic logs leads to interesting findings about trends in the Internet ecosystem, e.g. about content delivery networks and about the impact of IXP pricing models on the nature of peerings.

The discovery of IXP-based peerings is a related topic that has traditionally caught the attention of researchers for its implications in other areas, e.g. the study of the correspondence between IP addresses and ASes. Recent techniques (see, e.g., [8]) prove that most of such peerings cannot be found by simply mining traditional, “static” data sets like BGP routing data and Internet registry statistic files.

IXPs have also been studied [9], [10] as one of the causes of the evolution from a traditional hierarchical Internet to a more “flattened” version with AS-path getting shorter over time.

The renewed interest in the topic is confirmed by recent research work that aims at expanding the potential of traditional IXPs. Kim et al. [11] used logs of a large European IXP to study the status of IPv6 traffic during and after the World IPv6 Day in 2011 and the World IPv6 Launch in 2012. Gupta et al. recently introduced a prototypical software-defined Internet Exchange that enables expressive and highly customizable routing policies [12].

Despite of this large amount of investigation, a quantitative analysis on the impact of IXPs on key performance indicators like rtt, jitter, hop count, and packet loss is missing. In [13], the authors compare the rtt of a path between two points in the Internet when the path traverses an IXP and when it does not. Differently from us, their non-IXP path is simulated, by concatenating pieces of paths coming from several traceroutes. Namely, they find a common provider  $C$  of the source AS  $S$  and the destination AS  $D$ , and then find a  $S - C - D$  path in a traceroute graph computed from public traceroute datasets, such that an IXP is not traversed. Their conclusions are that the non-IXP paths have lower latency, which is in contrast with our results. In [14], the role of African IXPs is studied. Among other measurements, they perform traceroutes from 17 local probes towards fixed targets. They discover that many African ISPs do not have peerings at local IXPs, making local traffic paths often detour through Europe. They compute the improvement in rtt that would result by following a simulated path through an IXP. The simulation is done by replacing the detour part of a path with a direct link through an IXP or between pairs of local IXPs, for which they estimate the latency. Their conclusions are that African users would benefit from a higher connectivity between providers in African IXPs.

To our knowledge, all existing works that assess the effectiveness of IXPs by comparing key performance indicators use, at least partially, simulated data. The reason is that it is difficult to collect both a path traversing an IXP and one traversing only upstreams between a pair of hosts in a real-life network, where no direct control is possible. Supported by several ISPs, we fill this gap by comparing key performance indicators of real IXP and non-IXP paths.

### III. METHODOLOGY

The IXPs that are central to our study are the two oldest and most popular Italian IXPs, namely, MIX and NaMeX. MIX [15] is the largest in terms of peers: at the time of writing, its Website reports that 151 ISPs have peerings there, of which 106 flagged as “Italian”. We executed the procedure described in Section IV-A to determine the nationality of the ISPs declared to be Italian by MIX, and it confirmed the correctness of such declaration. NaMeX [16] currently serves 58 ISPs, as reported on their website. They are not classified based on their nationality, therefore we executed the aforementioned procedure and conclude that 47 of them are Italian. The number of peerings at MIX and at NaMeX is not publicly available. Executives at MIX kindly provided an estimation of more than 5,000 peerings in their IXP, while executives at NaMeX kindly disclosed an estimation of 438 peerings, i.e. 31.78% of the theoretical upper bound of 1378.

In our study we exploited the distributed infrastructure of RIPE Atlas [17], which is a Internet measurement network



Fig. 1. Maps showing the location of probes used in our experiments. MIX and NaMeX are marked with letters M and N, respectively.

based on thousands of devices, called *probes*, that are deployed in different types of networks (home, business, academic, etc) all around the world. Probes are connected to the Internet and can be remotely activated to schedule and perform standard network measurements (i.e. ping, traceroute, DNS resolution) towards arbitrary Internet targets. Each probe is labeled with its approximate location. We leverage RIPE Atlas to perform traceroute and pings from probes located in Italy and directed at a number of targets in Italy. The location of the probes used in our experiments is shown in Fig. 1. Several alternatives for distributed active network measurements are available nowadays. Examples include CAIDA’s Archipelago [18], Measurement Lab [19], and SamKnows [20]. However, the first two have very few probes in Italy, while the probes of the last are allocated to specific commercial targets and are not publicly available for research purposes.

#### A. Experiment CIS: Crucial Internet Services

In our first experiment we engineered network tests aimed at producing a plausible snapshot of the QoS associated to crucial websites commonly accessed by Italian users, with the goal of classifying the results according to the usage/non-usage of IXP peerings. As a first step, we selected two sets of Internet websites that can be considered crucial for Italian users. The first contains services related to critical infrastructures and the second contains popular sites. The first set, called CRITICAL, has been selected according to the taxonomy in [21] and contains 46 websites belonging to the following categories: on-line banking, insurance companies, public administrations, energy companies, legal courts, travel companies, health portals, and webmail providers. The second list, called VISITED, contains the 100 Italian websites most visited by Italian users, according to the ranking published by Alexa [22]. We mapped each domain name belonging to our lists to an IP address as follows. We scheduled ping measurements from all Italian probes targeted at each domain, once every 10 minutes for four hours and specifically forced probes to resolve the domain before each ping by querying their local DNS. We have then filtered the obtained IP addresses by keeping only those located in Italy according to the heuristic described in Section IV-A. The IP addresses obtained in this way are 53 for the first list and 94 for the second one. In the latter case we have fewer IP addresses than websites because some websites share the IP address. For each list, the test of EXPERIMENT CIS consists of 5 traceroutes, performed by

each probe at intervals of 10 minutes, for a total duration of 40 minutes. We intentionally performed few traceroutes in order to limit the total duration of the experiment, thus reducing the probability of measuring noise due to routing changes. Based on the collected traceroute data, for each traceroute, we computed four values: rtt, hop count, and two boolean flags telling whether MIX and NaMeX were traversed. To compute such flags we preliminarily collected all IP addresses used at the two IXPs for the establishment of peerings, by directly asking representatives of the IXPs. With such information at hand, we were able to flag all traceroute paths passing through either of the two IXPs.

### B. Experiment SBA: Selective BGP Announcements

In our second experiment we focused on discovering routing alternatives involving either IXPs or upstreams used by Italian ISPs to reach Internet services, in order to directly compare their performance. To do so we partnered with Mc-link, Seeweb and Unidata, three medium-sized Italian ISPs. In preparation for our experiment, we asked each partner ISP to reserve one IP subnet and one server in its data center. Each server was assigned an IP address falling within the reserved IP subnet. Also, it was configured to correctly handle and answer ICMP echo requests sent with pings and traceroutes. For each ISP we executed, at different times, the steps described below. We asked the ISP to announce a predetermined sequence of 5 BGP updates involving its reserved IP subnet. Each update had a lifetime of 4 hours, for a total of 20 nonconsecutive hours per test. The list of BGP updates was crafted to selectively distribute routes to different subsets of the available peers, adhering to the following scheme. 1) “UPSTREAM”: announce only to transit ASes. 2) “IXPS”: announce only to MIX and NaMeX peers. 3) “MIX”: announce only to MIX peers. 4) “NAMEX”: announce only to NaMeX peers. 5) “ALL”: announce to all peers. During each interval we ran traceroutes and pings from all Italian probes to the IP address of the server. For each interval we started one hour before the BGP announcement and ended one hour after the end of the interval. We performed one ping per minute and one traceroute every 10 minutes. The measurements ran before the first BGP announcement was performed, in order to assess the reachability and the performance stability of each target. The great majority of the intervals (not all of them because of technical constraints of one provider) were located during the daytime hours, to avoid any time-of-day effect on the measures [23]. Finally, at the end of each period, we filtered out all ping and traceroute measurements that took place near the 5 BGP announcements. More precisely, given the exact timestamp logs of BGP announcements, we identified 4 time intervals spanning one hour and centered at each of the timestamps, and removed each ping and traceroute falling within any of the intervals. We did so to account for potential instabilities and route flapping caused by the BGP announcements. For each probe-target-interval tuple, where an interval is one between “UPSTREAM”, “IXPS”, “MIX”, “NAMEX”, and “ALL”, we computed the average rtt, the average hop count, and the boolean flags. Observe that, while in EXPERIMENT CIS we took into account raw measured data, in EXPERIMENT SBA we considered average values. This was done to compare the performance of a probe-target pair measured in the different intervals. Further, from ping data, we

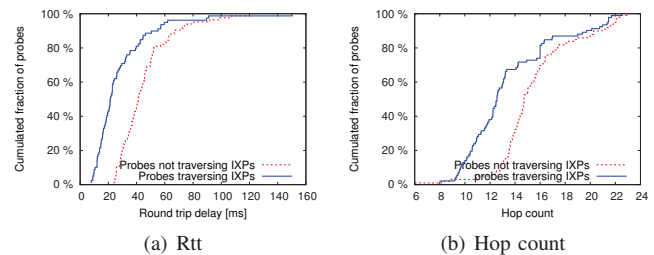


Fig. 2. Graphs comparing the performance of traceroutes that reach the targets of EXPERIMENT CIS traversing IXPs and those of traceroutes that use upstreams. (a) Targets from CRITICAL. (b) Targets from VISITED.

computed both the packet loss, measured as the sum of all the ratios between the number of packets that received no answer and the total number of sent packets, and the jitter, computed as the standard deviation of all rtt values. Unidata and MC-link have one data center each, located in Rome. Seeweb has two data centers, respectively located in Milan and Frosinone, so Seeweb reserved for us two servers, one in each data center and we targeted the measurement to both.

## IV. ANALYSIS

### A. Experiment CIS

The actual number of probes that were used for EXPERIMENT CIS is 91 for the measurements towards targets in dataset CRITICAL, and 104 for the measurements towards targets in dataset VISITED. Figure 2 contains graphs that give a high level overview of how performance, as measured by Atlas probes, depend on the presence of IXPs in the traceroute paths. The performance indicators used to generate the graphs are the rtt and the hop count, respectively. Each graph contains two plots, one for all the traceroutes that use paths traversing one of MIX and NaMeX (blue) and the other for traceroutes that do not see any of the two IXPs in the traceroute paths (red). Each plot is a CDF showing the distribution of measured values for the selected performance indicator. Fig. 2(a) is for dataset CRITICAL, while Fig. 2(b) is for dataset VISITED. For brevity we omit the figure on rtt of VISITED and the figure on hop count for CRITICAL which exhibit exactly the same pattern. The plots show that, in general, probes traversing IXPs have better indicators. For example, referring to Fig. 2(a), about 70% of probes that choose IXPs have average rtt of 30ms or less, while only 20% of those that do not traverse IXPs have the same performance. However, such results must be considered very carefully, because of the following three reasons: 1) The two CDFs of Fig. 2(a) (the same holds for Fig. 2(b)) refer to disjoint sets of probes, since each probe either passes through an IXP or not. In principle, it is possible that a probe currently passing through an IXP could have a better performance when forced, in some way, to pass through an upstream (or vice-versa). 2) Rtt and hop counts refer to the last hop that replied to our measurements, which quite often is not the actual target (40 cases out of 53 for dataset CRITICAL, and 59 cases out of 94 for dataset VISITED). 3) A router in an IXP might answer to a traceroute using an interface that is not on the peering lan, leading to a misclassification of the corresponding traceroute. Also, the data on hop count could be affected by the presence of tunnels (e.g. transparent MPLS tunnels [24]). On the other hand this might happen both to packets that traverse IXPs and

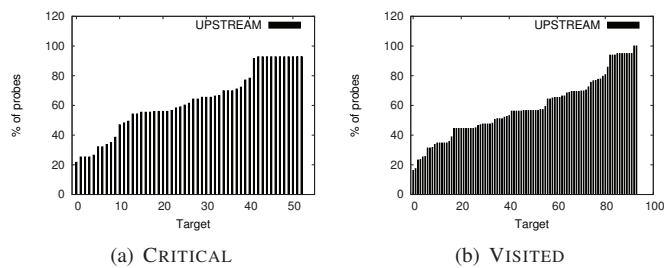


Fig. 3. % of probes that reach a target with an upstream in EXPERIMENT CIS.

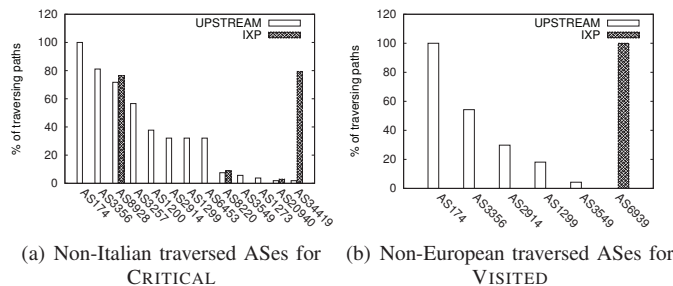


Fig. 4. % of paths traversing foreign ASes when an IXP is used or not.

to packets that do not. However, these graphs help us draw a first positive impression on the impact of IXPs. Observe that EXPERIMENT SBA does not suffer of the problems pointed out in items 1–3.

Next we study how Atlas probes distributed their traffic between paths including or excluding IXPs during EXPERIMENT CIS. Figure 3 shows, for each target, the percentage of probes that reached it through an upstream. These graphs show that most targets are reached preferably through upstreams. In particular, in Fig 3(a), 40 targets over 53 were reached through an upstream by more than 50% of probes, and, in Fig 3(b), 60 targets over 94 were reached through an upstream by more than 50% of probes.

Finally, the data collected in EXPERIMENT CIS can be analyzed in terms of security issues associated with the traffic targeted at crucial Web services. Indeed, recent cases of network traffic espionage [4] solicited governments to check if the traffic generated by their citizens and targeted to critical Web services remains or not inside the countries [5]. It is perfectly clear that this type of locality does not give security guarantees on such a traffic, but it is equally clear that, nowadays, governments are very sensitive to this issue. Therefore, we measure how frequently using an IXP to reach an Italian target from an Italian source permits to avoid the transit through a foreign upstream. That is, we quantitatively check whether MIX and NaMeX are effective in keeping the local traffic local. In order to determine the country of an AS  $X$ , we query the RIPEStat [25] service asking for the prefixes announced by  $X$ , along with the country of each of these prefixes. If the returned prefixes are all associated to the same country, then we assume that it is the country of  $X$ . Otherwise, to determine its country, we manually retrieve metadata on  $X$  looking, e.g., at the website of the corresponding provider. Fig. 4 illustrates our results in terms of locality. The statistics are computed as follows. First, the ASes traversed by paths in EXPERIMENT

CIS are labeled as *non-Italian* and/or *non-European* if the nation detected by the aforementioned heuristic is non-Italian and/or non-European. Also, the source-target pairs of the experiment are split into two groups, based on the presence of IXPs in their paths. Finally, each detected non-Italian and non-European AS is associated with the source-target pairs that traverse it. Fig. 4 shows the results of the computation. As an example, abscissa 3356 of Fig. 4(a) shows that 80% of the source-target pairs not passing through an IXP pass through AS3356 (that is not Italian) while no one of the source-target pairs passing through an IXP traverse it. In most cases, a foreign AS is traversed only when an upstream is used, or in great prevalence when an upstream is used, which confirms that IXPs are effective in preserving the locality of traffic. A few exceptions are ASes 8928 (INTERROUTE), 8220 (COLT), 34419 (VODAFONE), 20940 (AKAMAI), and 6939 (HURRICANE). We have detected, by checking the traceroute paths, that INTERROUTE, COLT and HURRICANE have peerings at IXPs, so that is the reason for their high values in the IXP cases of Fig. 4. VODAFONE, instead, receives traffic from the Italian Vodafone AS, so it is a case of traversed foreign AS that cannot be avoided by using IXPs. AKAMAI is an interesting case, since in our experiments it receives traffic from BT Italia, which contains one of the targets of EXPERIMENT CIS, and then sends it back to the same AS. Our guess is that AKAMAI, as a content provider, has some network infrastructure inside the data centers of BT Italia, and it possibly hosts some targets of our measurements.

### B. Experiment SBA

For EXPERIMENT SBA we started by selecting the subset of probes, called multi-choice, that were able to reach the target during both the “UPSTREAM” interval and during at least one of the “MIX” and the “NAMEX” intervals (see Section III-B). These probes allow to measure the differences between the connectivity offered by IXPs and that offered by upstream providers. The actual number of multi-choice probes amounts to 83 for the Seeweb Milan target, 81 for the Seeweb Frosinone target, 74 for MC-link, and 80 for Unidata. Fig. 5 shows the distribution of performance indicators on all multi-choice probes. Fig. 5(a) shows the roundtrip delays measured during the experiment with Seeweb, Fig. 5(c) shows the hop count measured during the experiment with MC-link, and Fig. 5(e) shows the jitter measured during the experiment with Unidata. Because of space limitations we omit figures showing other combinations of performance indicators and ISPs, since they show analogous trends.

As an example, Fig. 5(a) contains 4 plots that show the distribution of the average rtt for the two Seeweb targets, in both cases when the target was reached through an IXP or through an upstream. Consider that in this and in all the following figures the plots involving paths through IXPs show the “best” option, i.e. for each performance indicator and for each probe that reached the target during both the “*MIX*” and “*NAMEX*” intervals, it is shown the smallest value between the two options. Observe that about 50% of multi-choice probes reach both targets with an average rtt of at most 15ms when using IXP connectivity, compared to only about 30% of them in case of upstream connectivity. The measurements towards the Milan target and the Frosinone target exhibit comparable performance.

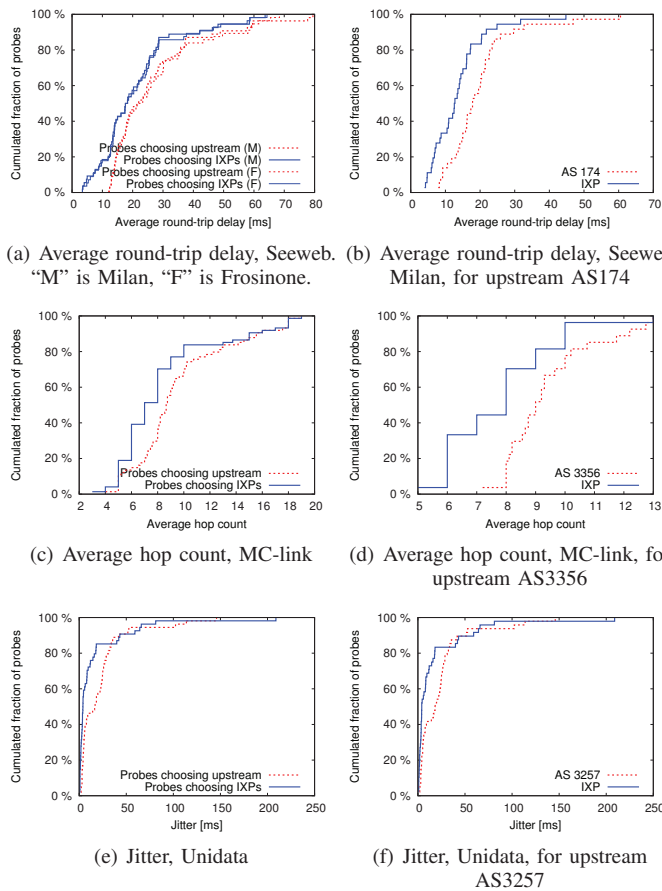


Fig. 5. Comparison of performance indicators between upstream and IXP connectivity, as measured in EXPERIMENT SBA. For each of our three partner ISPs, one performance indicator is shown. The same data is also plotted restricted to a specific upstream.

Figs. 5(a), 5(c), and 5(e) put in evidence that paths traversing IXPs always give performance that are better or at least equal to those traversing upstreams, for each provider among Seeweb, MC-link, and Unidata, and for each considered performance indicator among average rtt, average hop count, jitter, and packet loss. The latter is the only metric for which IXPs and upstreams exhibit about the same performance, with negligible differences, and its graphs are omitted for brevity.

During the experiments we noticed that paths through the upstreams exhibited quite different values of performance indicators depending on the specific traversed upstream. So, looking at the traceroutes gathered during the “UPSTREAM” interval, we found the upstreams of each of our providers and deepened the analysis for all of them. Here is the list of upstreams that we found: 1) MC-link (AS3257, AS12874, AS174, AS3356, AS35612, AS57329); 2) Seeweb (AS3257, AS174, AS3549, AS3356); and 3) Unidata (AS3257, AS12874, AS16004, AS24796, AS20836). Figs. 5(b), 5(d), and 5(f) show the main outcome of the analysis. As an example, Fig. 5(b) contains the same data of Fig. 5(a) restricted to the Milan target and to the probes that reached the target through AS174 (COGENT). The IXP plot is restricted to those probes too. Observe that the plot has exactly the same trend of Fig. 5(a). It follows that the paths traversing AS147 are the principal factor that makes the average upstream performance

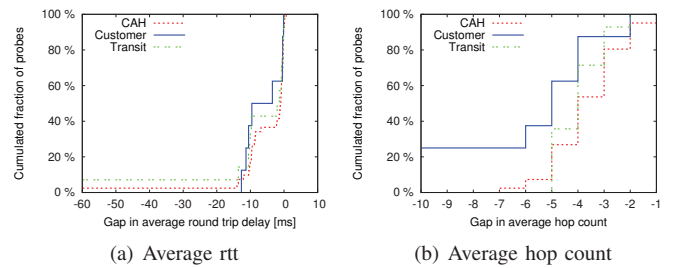


Fig. 6. Difference in performance based on the AS classes of the probes.

lower than the IXP performance. Similar arguments apply to Figs. 5(d) and 5(f). In the first one, AS3356 (LEVEL3) is shown as the upstream with the largest difference from the IXP alternative among all the upstreams of MC-link. In the second one, AS3257 (TINET) exhibits a behavior similar to the plot of Fig. 5(e). We argue that the performance obtained from upstream connectivity heavily depend on the choice of a specific upstream.

As a next step, we consider the classification of ASes in [26]. Our goal is to identify different trends in the improvement of performance indicators based on the type of AS that hosts a probe. ASes are assigned to 3 classes: 1) *Customers* represent various organizations, universities and companies at the network edge that are mostly users of the network. 2) *Transit* are ISPs that provide Internet access and transit services. Transit aim to maximize their customer base in their geographical area and to reduce their upstream transit costs through selective peering with ISPs. 3) *Content/Access/Hosting Providers* (CAH) are ISPs that offer Internet access and/or server hosting. Their access customers can be residential users or enterprises with no AS numbers, while their server hosting customers are content/service providers with no AS numbers. Figure 6 presents two graphs, respectively based on average rtt and average hop count for the Seeweb Milan target during EXPERIMENT SBA. Each graph contains 3 plots, each corresponding to one AS class. A plot is a CDF that shows the distribution of performance gaps derived from the comparison of performance indicators between paths traversing IXPs and paths avoiding them, as seen by each probe. Graphs show that Customer ASes hosting probes in Italy are the ones that benefit the most from the availability of IXPs. For example, about 50% of probes in Customer ASes see an improvement in the average rtt of at least 10ms when traversing IXPs, as opposed to about 20% and about 40% for probes in CAH and Transit ASes, respectively. Similarly, about 62% of probes in the first class has path lengths reduced by at least 5 hops, while the other two classes only reach about 25% and about 38%.

Next, we leverage data collected during the execution of EXPERIMENT SBA to understand how many probes picked the “wrong” path to the destination when all options were available (i.e. during the “ALL” interval), based on the comparative analysis of performance indicators. By “wrong”, or “incorrect”, we actually intend that a path is suboptimal from the point of view of one of the performance indicators we have considered, while a different path would have had better performance. Obviously, an ISP may have motivations that go beyond our metrics for choosing a path instead of another. Fig. 7 presents two graphs based on average rtt and average hop count,

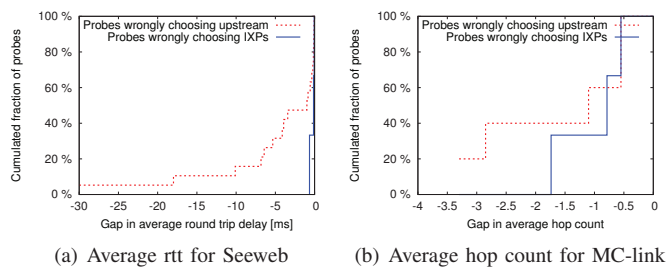


Fig. 7. Comparison of routing choices made in the “ALL” phase of EXPERIMENT SBA that lead lower performance than alternative routing choices. The measurement are performed towards (a) Seeweb target in Milan and (b) MC-link target.

measured respectively towards the Seeweb Milan target and the MC-link target. Each graph contains two plots, one for probes that incorrectly preferred upstreams over IXPs (in red), and one for probes that incorrectly preferred IXPs over upstreams (in blue). Each plot is a CDF that on the  $x$  axis has performance gap values for a given metric, computed as  $metric_a - metric_c$  where  $c$  and  $a$  are respectively the current path traversed by a probe and the alternative path that was available. One between  $c$  and  $a$  traverses an IXP, while the other does not. Only negative gaps are shown, so the graphs represents those probes that made a “wrong” choice between traversing an IXP or an upstream to reach the target. For a given performance gap  $g$ , on the  $y$  axis there is the percentage of probes with a gap less or equal to  $g$  over all probes with a “wrong” choice. More informally, the CDFs gives a quantitative overview of “wrong” choices based on how much the performance indicators would really improve by switching routing options (i.e. traversing IXPs for the red curve, and avoiding them for the blue curve). The analysis of both graphs reveals that a relevant percentage of “wrong” upstream transits actually lead to non-negligible performance degradations, whereas “wrong” transits through IXPs generally correspond to much less negative effects. For example, referring to Fig. 7(a), about 30% of probes that choose upstreams would reduce their average rtt by at least 5ms by switching to an IXP, while no single switch in the opposite direction would give the same tangible effect.

## V. CONCLUSIONS AND FUTURE WORK

We investigated the role played by IXPs in the Italian Internet ecosystem. Our experiments put in evidence that peerings exploiting IXPs have a positive effect on key performance indicators such as latency, hop count, packet loss, and jitter. Also, they reduce the number of foreign ISPs traversed by the traffic between Italian citizens and critical Internet services like Banks and Public Administrations. We conclude that if an ISP motivates a de-peering through an IXP in terms of performance improvements it makes a statement that is at least questionable.

It would be interesting to extend our experiments considering more performance indicators. Unfortunately, the measurement network used in our experiments, although technically ready for measuring more parameters (e.g. for bandwidth measurement), adopts a conservative approach to limit the impact of experiments on users that host probes. Further, we plan to cooperate with ISPs and IXPs in other countries to reproduce the experiments, compare the results, and generalize

the methodology. Another direction for extending our experiments would be to consider full QoE issues. However, this would imply application level measurements which are even harder to perform on common measurement networks.

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